

MIDDLE EAST TECHNICAL UNIVERSITY

NORTHERN CYPRUS CAMPUS

DEPARTMENT
OF
COMPUTER ENGINEERING

CNG 351
Data Management and File Structures
Assignment 1

(4% of actual grade)

DUE DATE: 29, October 2026, 23:55 (Cyprus Time)

IMPORTANT RULES

1. Use a graphical tool such as Lucidchart or Draw.io.

   - Lucidchart can be used to design and create your solution. You can find a full tutorial on ER diagrams and Lucidchart here: `https://www.lucidchart.com/pages/er-diagrams`. Please make sure that you use **Chen notation** for your design, other notations will not be accepted.

     For Lucidchart, please ensure that you have an educational account. **To get an educational account, you need to sign up with your METU mail addresses from the following link:** `https://lucid.app/pricing/lucidchart`

   - Draw.io can also be used to draw your diagram. You can find the full details here: `www.draw.io/`.

2. Any assumptions must be explicitly explained. If your assumptions do not make sense in the given requirements content or conflict with the requirements, or similarly, if you assume and do not specify the assumption, you will lose points from that part of your design. Please do not include trivial assumptions; for example, "we assume id is unique".

3. Create a PDF file for your solution and upload only one PDF file to ODTUClass, one team member is enough to upload the solution.

4. The assignments will be completed by a team of two/three people. Your team details are recorded on odtuclass. Please note that you **cannot** change your team members.

5. Please submit a report that includes a cover page with the team details, including your team/group Id, group/team member names/surnames and their student IDs.

6. Finally, this assignment submission is subject to the "professionalism and Ethics" rules given in the syllabus. Therefore, we may conduct an oral or written test to confirm students' knowledge and assess their grades. If the follow-up exam (oral/written) results in failure, then we can apply the rules written in the course syllabus.

PURPOSE

The main purpose of this assignment is to review the modelling of a relational database from requirements using Entity-Relationship Diagrams (ERD).

GRADING

This assignment includes four use cases for different database applications. The data requirements of each of these use cases need to be modelled with ERD, and the grading will be as follows:

1. Use Case 1: 30 points;
2. Use Case 2: 32 points;
3. Use Case 3: 33 points;
4. Please note that you are not designing the application; you are designing the **data requirements** of the application.
5. Your designs are expected to be clean and readable. You will also be penalised if you use ternary (or n-ary) relations. Please make sure that all your designs consider cardinality and participation constraints.
6. Please make sure that you use **Chen notation** for your design, other notations will not be accepted and you will receive zero from your design.
7. Report: 5 points. A good report means typewritten, complete (every section fulfilled), with clear explanations in English (where relevant), and submitted via ODTUCLASS as one combined PDF document. The report's first page must be a title page that clearly states team details and assignment number. Each use case should also include *Assumptions* written. The footer of all subsequent pages should be numbered in the format x of y (e.g. 2 of 6), etc.

USE CASES

Use Case 1 - WHO WANTS TO BE A MILLIONAIRE?

"Who Wants to Be a Millionaire?" is a famous television quiz show where contestants answer multiple-choice questions and try to increase their number of correct answers to win more money. Because many episodes are recorded and many different contestants and questions are involved, the television company wants to create an application to manage the show. Your task here is to design the database for that application in such a way that it captures the data details described below.

The quiz show is broadcast in separate episodes. Each episode is assigned a unique identifier, and information such as the episode number, recording date and broadcast date is recorded. For an episode, the system also records its season. Please note that in a season, at most 10 episodes can be produced and the system keeps track of the year of production of a season.

Several contestants may participate in the same episode. For each contestant, the system records personal information such as full name which is composed of name and surname, date of birth, occupation, city and phone number. The contestant's age is automatically determined from the date of birth. Please note that the system records the contestant details once they participate to an episode and a contestent can only participate to this show once.

The system also maintains a collection of multiple choice questions which can be used during the show. Each question has a unique question ID, question text, four choices of answers, and the correct answer and a difficulty level. The difficulty level shows how difficult a question is, such as easy, medium or hard.

Questions are also organised into categories such as history, science, sports, geography and entertainment. A question may belong to several categories at the same time. For example, a question may be related to both sports and geography. Similarly, each category may include many different questions. Every question should be associated with at least one category.

During the competition, contestants answer questions one by one. A contestant may answer several questions and, of course, the same question may be asked to different contestants over time. Every time a question is asked to a contestant, the system should record the answer given by the contestant and whether this answer was correct or not. A contestant cannot complete their participation without being asked at least one question.

In addition, contestants may use jokers while answering questions. The available jokers are Fifty-Fifty, Phone a Friend, or Ask the Audience. The system records a short description of the jokers and the date and time when they use it. Every time a contestant uses a joker, the system must also record which question it was used for. Please note that jokers cannot be recorded without contestant details. However, the contestants can complete the game without using any jokers, and they can use at most four jokers.

Finally, hosts present the episodes. For each host, the system records personal information such as full name (first name and surname), phone number, and unique email address, along with a unique host ID. Each episode must have a host, while a host may present many different episodes over time.

Use Case 2 - CleanCampus

At METU Northern Cyprus Campus, hundreds of students live in dormitories and rely on shared laundry machines. A company called CleanCampus has decided to develop a centralised platform to efficiently manage laundry facilities, users, equipment, and related operations across the campus. Your task here is to help design the

database for this application.

First, CleanCampus must manage washing machines across the campus. The machines are located in shared laundary rooms and they serve several dormitories. Each machine has a unique ID, label, type which is standard or industrial, installation date, and warranty end date. The brand of the washing machine is also recorded (e.g., Bosch, Samsung, Miele, etc.). Before a dormitory's residents can use a machine, the machine should be allocated to the dormitory. Each dormitory has a unique ID, dorm name, campus zone, and level (undergraduate or graduate). A dormitory can be allocated many machines, and a machine can serve many dormitories. For each allocation, the system records the usage rank and the free credit amount given to the dormitory. The system ranks dormitories by how heavily they use each machine, with the top dormitory receiving special recognition as the leading user of that machine. This status should be determined automatically from the recorded usage rank.

The system also manages the machine models on which machines are based. In the system, each model is recorded with a unique ID, model name, capacity information and supported programs. A typical model supports multiple programs such as cotton, wool, quick wash, or delicate. The capacity information also includes maximum and minimum load. In the system only one model can be recorded for a machine.

A dormitory can accommodate multiple students at the same time. Each student has a unique student id, a unique username, name, surname, and age. Every student is currently assigned to one dormitory, and each dormitory can have multiple students assigned to it. The system also records the history of students moving between dormitories. Whenever a student moves to a different dormitory, the system records the dormitory involved, the date the student started staying there, and the date the stay ended. A student may stay in the same dormitory again after living in another dormitory. Over the years, a dormitory can accommodate many different students, while a student can be associated with several dormitories throughout their time on campus.

Each machine can have multiple usage sessions. A usage session is identified by a session number that is unique only within the machine on which the session occurs. Therefore, the same session number may be used for sessions belonging to different machines. Every session is associated with exactly one machine. For each session, the system records the session number, date, start time, and end time. The session duration is calculated automatically from the start and end times. Sessions may be scheduled, completed, cancelled, or interrupted, so the system also records each session's status. When a session takes place, the system should also record the dormitory.

In this system, suppliers are also a vital part of the service. Each supplier has a unique ID, name, and specific industry, such as spare parts, detergent, or maintenance. A supplier can maintain multiple machines, and each machine can have

multiple suppliers. CleanCampus does not maintain the machines itself, but it has contracts with suppliers for specific machines. The system tracks maintenance contract details, such as the start date, end date, and contract amount. The contract must clearly indicate the supplier and the machine it covers. Of course, a machine can have many contracts associated.

Use Case 3 - METU NCC KALTEV

Kalkanli Technology Valley (KALTEV) is the technology park of METU Northern Cyprus Campus. It is one building where companies and startups rent offices, work on research and development (R&D) projects, hire staff, take student interns and meet investors. KALTEV management wants an online system to manage its daily operations and activities in the building, and your task here is to design the back-end database for this application.

- Every company in KALTEV must be registered in the system. For each company, the system keeps its name, tax number, founding date, sector (for example, software, energy or biotech), website and contact information. Contact information includes an email address and a phone number. Two companies can never have the same tax number. Moreover, management often wants to see how many years each company has been operating (the company's age). There is no need to enter this information explicitly to the system, and this can be worked out from the founding date.

- KALTEV has many offices, and this system should allow management to track the rental details of these offices. Each office has a unique office number, floor, and size in square meters. A company cannot work in KALTEV without an office, and a growing company may rent more than one office at the same time. For each rental, the system keeps the start date, end date, and monthly rent. When a rental ends, the office can be given to another company, but KALTEV still wants to know which companies used which offices in the past. Some offices may be empty.

- KALTEV management wants to know who is currently working in the building. Therefore, companies add their employee details to the system. For each employee, the system stores their unique national ID, first name, surname, unique email, job title, and phone numbers. Most employees give a mobile and an office number, and some give more. A person works for only one KALTEV company at a time, and the system keeps the date they started there. Each company chooses one of its own employees as its official representative. This person signs all official papers with KALTEV, and nobody can be the representative for two companies. The system records employees for each company, but KALTEV management must easily access the number of people working for a company; therefore, the system automatically calculates and stores this for each company.

- In KALTEV, companies can run R&D projects, and the system must also track these projects. Each project has a unique project code, a title, a budget, a start date, and an end date. Projects can also be in the following stages: proposed, active, completed, or cancelled, and the system must track this. Projects cannot be shared in KALTEV; every project belongs to one company, and a company may run many projects. Each project is divided into parts called deliverables, named D1, D2, D3, and so on. Every project starts counting its deliverables from D1, so many projects have a D1 deliverable. For each deliverable, the system keeps its number, title, due date, and the date it was actually submitted. A deliverable has no meaning without its project.

- Companies in KALTEV also need to record which employee works on which projects. Employees work on projects, and the system keeps how many hours per week each person works on each project. One person can work on several projects at the same time. Each project has a dedicated project manager, who is chosen from the people working on that project.

- KALTEV also allows students to be interns and the management would like to keep track of those interns. For each student, the system keeps their student ID, first name, surname and department, plus the start and end dates of their internship. KALTEV only keeps a student's current internship. When an internship ends, the system removes the student's record. Every intern has a supervisor from the company where they are doing the internship. One employee may supervise several interns.

- The KALTEV building also has ten shared meeting rooms. Each meeting room has a unique room number, floor, size in square meters, and capacity. Companies can book meeting rooms, and for each booking the system stores the date, start time, and end time. A room can never have two bookings that start on the same date at the same time, but two different rooms can of course be booked at the same time. KALTEV management also wants to know which company booked the room; otherwise, it cannot process the booking.

- KALTEV is a technology park, so companies always seek funding from investors. Management also wants to maintain investor information in the system. For each investor, KALTEV stores their name, address and an email as contact details. An investor can invest in several companies, and a company can receive investment from several investors. Some companies have no investors yet. For each investor and company, the system keeps the date and the amount. Therefore, management wants to know how much each investor has invested in a company and the total investment received so far.